Steve Morales

Software Engineer | Test Automation · Python · C++ · CI/CD
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github.com/Steve-Morales | US Citizen in San Diego, CA

Skills

Languages: Python, JavaScript, C++, Node.js, HTML, CSS, Bash

Testing & CI/CD: test automation, Google Test, Jenkins, GitHub Actions, Docker, GDB, Valgrind, Jira

Tools & AI: Git, Linux, React, FastAPI, PostgreSQL, RESTful services, AI-assisted development (ChatGPT, Gemini)

Education

M.S. in Software Engineering, Arizona State University, GPA 4.0 — August 2025 – May 2027 (Expected)

B.S. in Computer Engineering, University of California, San Diego — Graduated

Experience

Software Developer, Tactical Engineering and Analysis, CA — *March 2025 – May 2026*

- Expanded test coverage for a real-time vehicle control system used by operators by developing 100+ automated Google Test cases for 5 C++ interfaces in a Jenkins CI/CD pipeline.
- Reduced network bandwidth by about 80% so operators could command 8+ vehicles at once by leading a new interface design, tested with Python test scripts and a SWIG mocking layer.
- Eliminated a system-wide slowdown, reducing disk reads by over 90% in iostat, by troubleshooting a C++ service with GDB and Valgrind and replacing its polling with a Linux inotify observer.
- Repaired wrong map locations shown to operators by tracing a client-server data route from the UI through Linux logs and correcting the backend packet route with the backend team.
- Streamlined daily development for a 7-person agile team with 3+ Python automation scripts, built faster with AI-assisted development tools (ChatGPT, Gemini) for coding on Linux.

Junior Software Engineer, Programs Management Analytics & Technologies, CA — *July 2023 – March 2025*

- Verified a multi-vehicle simulation platform under a tight deadline by executing 25 software test procedures from Jira on Linux and non-PC lab hardware (human-machine interface).
- Created a Bash script that automated Docker and Linux test environment setup, documented in 2 setup guides used to onboard engineers so the team stopped losing time to manual configuration.
- Resolved defects found during end-to-end testing by triaging Linux logs with grep and tail, then submitted fixes as Git merge requests, applying subject matter expert (SME) feedback.

Projects

Proctor, AI Agent Testing Platform (Personal Project) — *June 2026 – Present*

- Built Proctor, a test automation platform that evaluates artificial intelligence (AI) agents on simulated websites, as 3 Docker services with a GitHub Actions CI/CD pipeline in Git.
- Designed React and JavaScript test websites that reset to the same seeded state every run so each AI agent faced identical screens and pass or fail results stayed repeatable.
- Engineered RESTful services in Python with FastAPI so testers could create, assign, and track AI agent test tasks, storing every agent action in PostgreSQL for failure triage.

Team Lead, M.A.R.V.I.N. Medical Visualization Project — *February 2023 – May 2023*

- Led a team of 4 building a medical augmented reality app that streamed live ultrasound video to a HoloLens 2 headset via a Node.js server and a JavaScript, HTML, and CSS desktop app.
- Consulted with the physician sponsor, a non-technical stakeholder, translating surgeon feedback into technical requirements so the app fit how operating room teams actually worked.